Action Classification & Spotting in Videos

Using the SoccerNet Dataset

Team 7

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***Module**: C6 - Video Analysis*
***Coordinators**: Albert Clapés and Artur Xarles*

Presentation overview

1 Introduction
- Objective
- Dataset

2 Insights from past weeks
- Week 5
- Week 6

3 Best-performing model
- 3.1 Successful approaches
- 3.2 Unsuccessful approaches

4 Impact of time aggr.
- Multiple tolerances
- Comparison

5 Further model analysis
- Visualizations
- Qualitative analysis

6 Discussion & conclusions
- Challenges
- Future work

1

Introduction

Objectives

Ball Action Classification

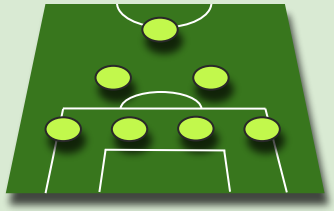
Classify soccer actions within a clip

Ball Action Spotting

Spot soccer actions precisely

SoccerNet[1]

4-2-1



~15% Testing

~25% Validation

~60% Training

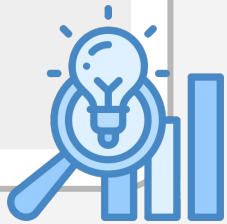
12 ball actions

- Pass
- Drive
- Header
- High Pass
- Out
- Cross
- Throw In
- Shot
- Ball Player Block
- Player Successful Tackle
- Free Kick
- Goal

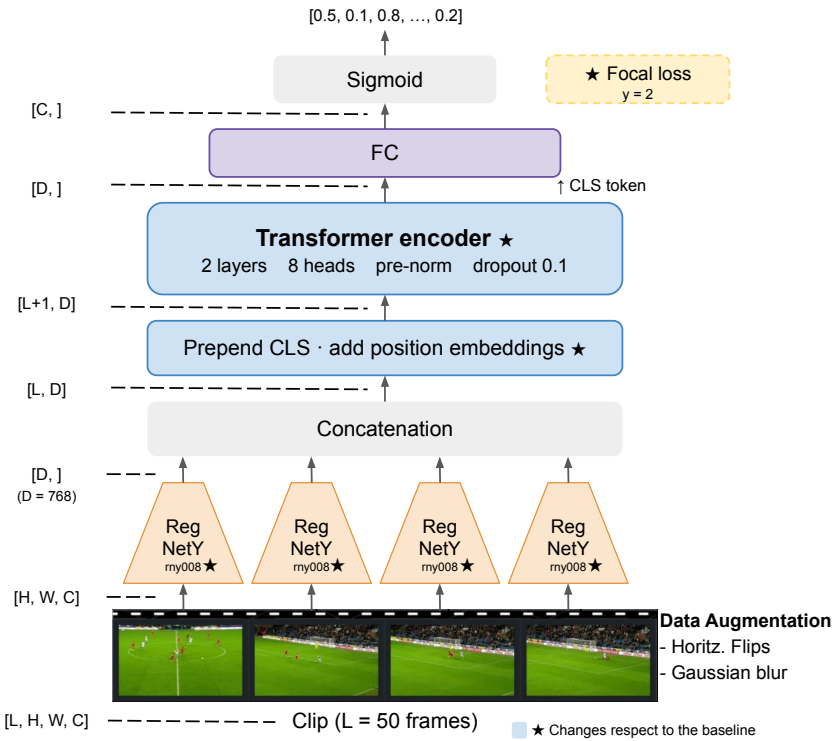
1. Giancola, Silvio, et al. "SoccerNet: A scalable dataset for action spotting in soccer videos." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2018.

2

Insights from past weeks



Week 5 – Action Classification



49.79 mAP10↑	**51.17** AP12
15.00 M parameters	**71.45** GMACs

Main changes:

- Backbone change from RNY004 to RNY008
- Transformer instead of maxpooling.
- Focal loss.

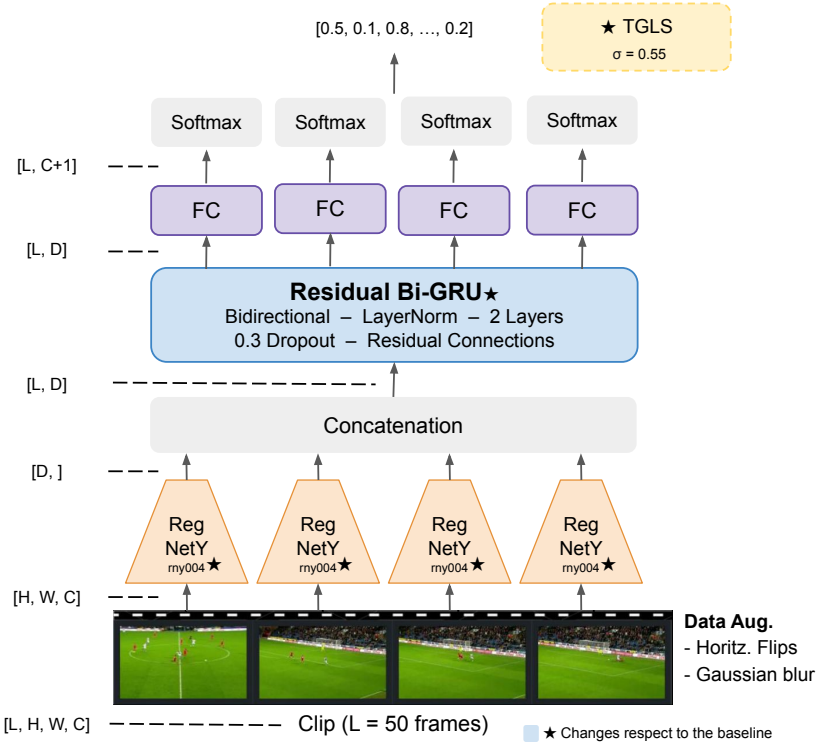
Takeaways:

- Trainable backbone better than frozen.
- Temporal modelling makes a different but many architectures are hard to train.

Challenges:

- Sparse ball actions in long clips.
- Background frames dominate the signal.

Week 6 – Action Spotting (without temporal aggregation)



33.79 mAP10↑	**30.43** mAP12
5.66 M parameters	**36.3** GMACs

Main changes:
- Backbone change from RNY002 to RNY004.
- Residual Bi-GRU.
- Temporal label smoothing (Gaussian).

Takeaways:
- The backbone is as important as the temporal modelling that comes next (similar gains).

Challenges:
- Visually ambiguous classes.
- Easy to overfit.

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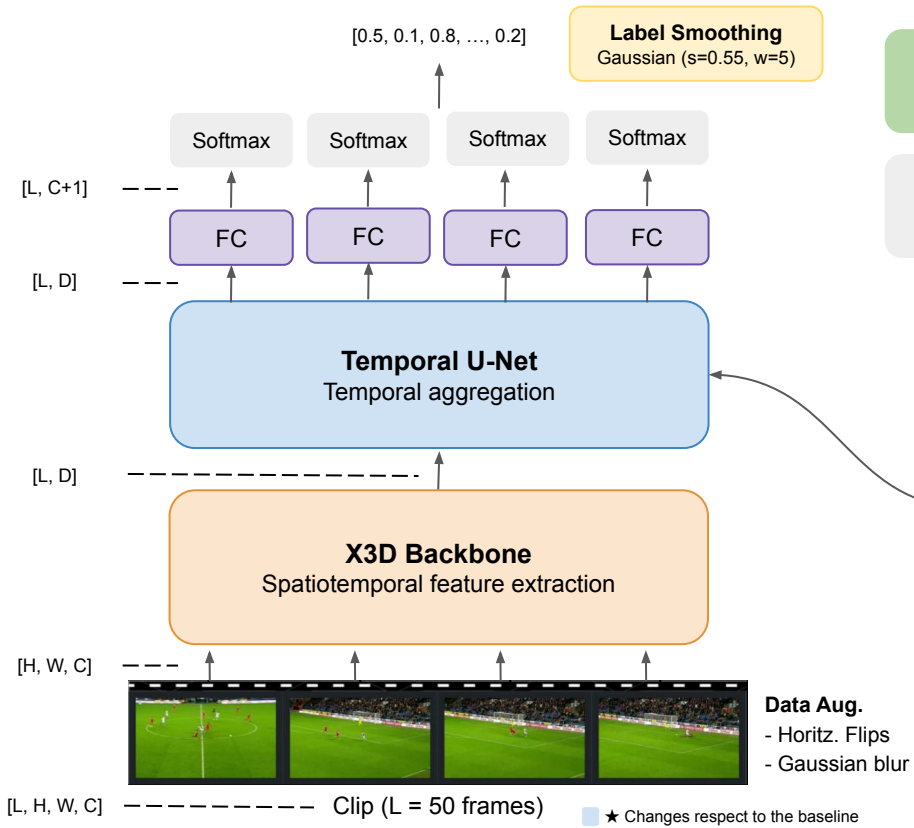
Best model overview

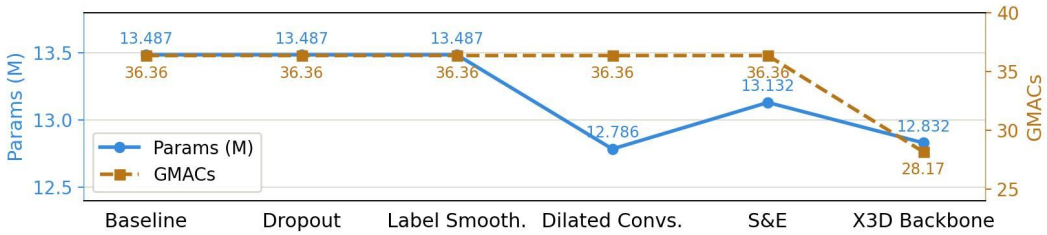
3 Best model overview

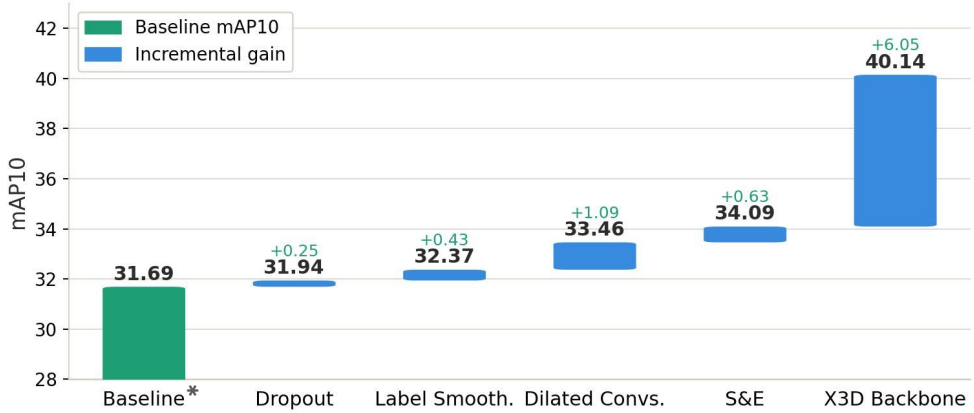


40.14 mAP10↑	**35.73** mAP12
12.8 M parameters	**28.2** GMACs

- 3 encoder & 3 decoder layers
- Temporal dimension reduced to 6
- Dropout
- Dilated convolutions
- Lightweight attention (squeeze and excitation blocks)

Main ablations





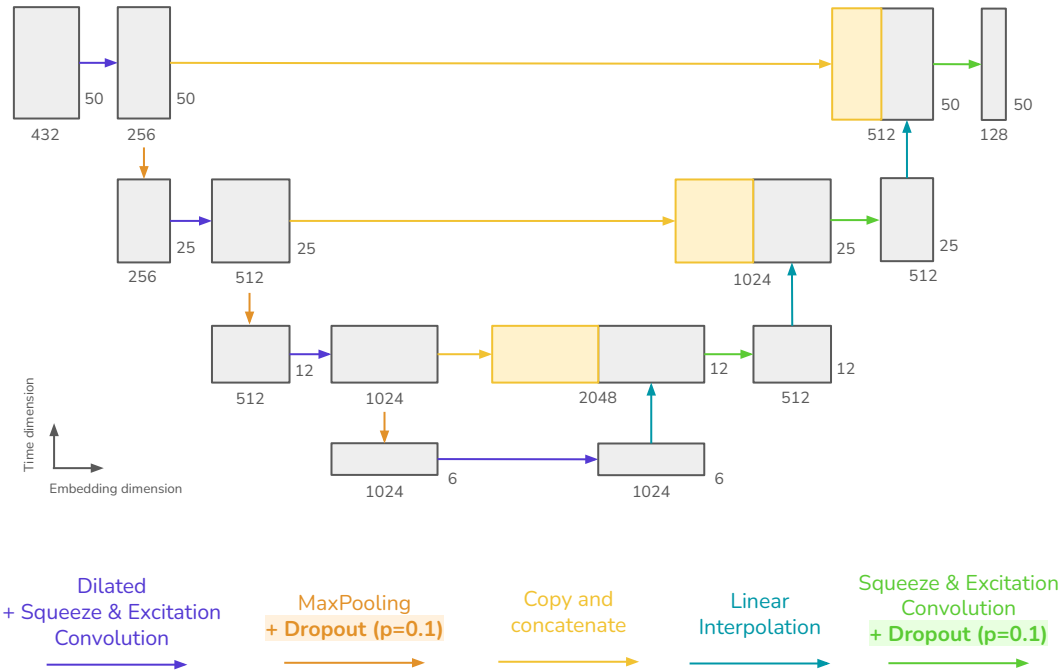
Sequential ablations to measure isolated impact on mAP10@1s

Motivation behind each step:

- **Dropout:** regularization to reduce overfitting
- **Label smoothing:** avoid unfairly penalizing the model temporally.
- **Dilated convolutions:** capture multi-scale temporal patterns
- **Squeeze-and-Excite:** amplify informative channels and suppress noisy ones
- **X3D backbone:** capture spatio-temporal patterns at the backbone directly

***Baseline** consists of RNY004 + U-Net with vanilla convolutions

Dropout



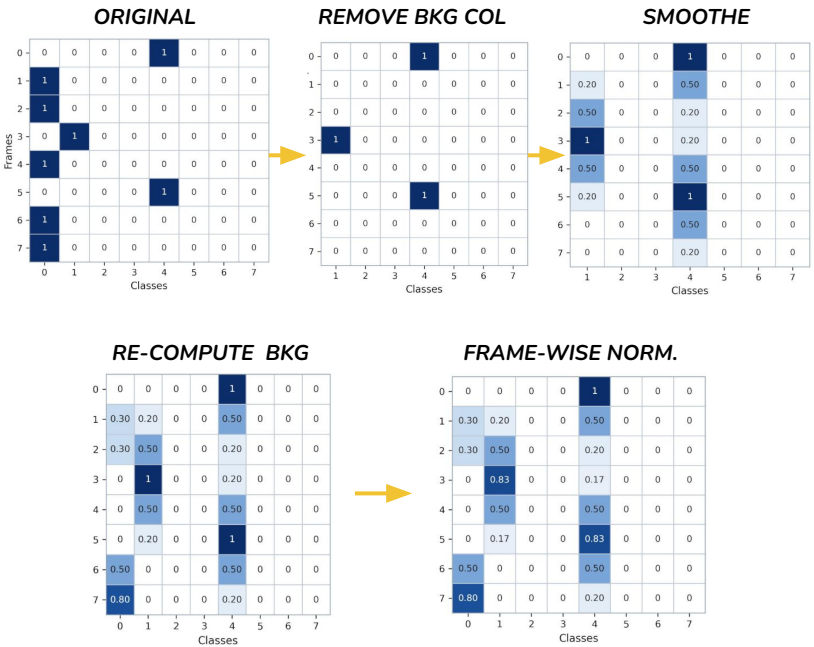
+0.25 (+0.8%)
in mAP10

Comparison with previous stage

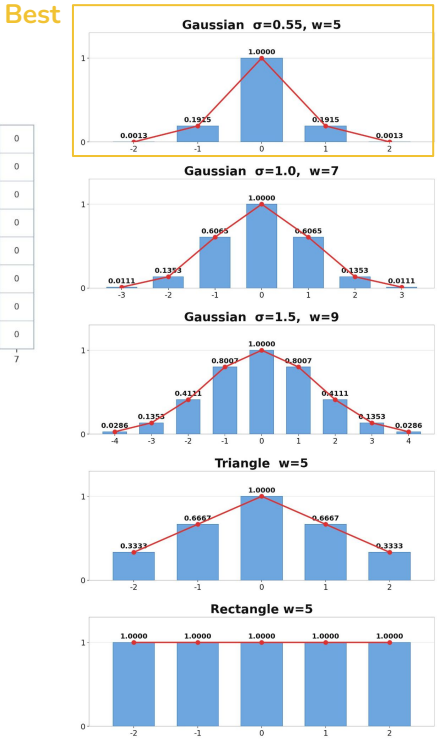
- Applied at every layer.
- Low probability to avoid training collapse.
- Only experimented with this probability value.

Temporal label smoothing

Label smoothing steps (toy example):



Different smoothing kernels

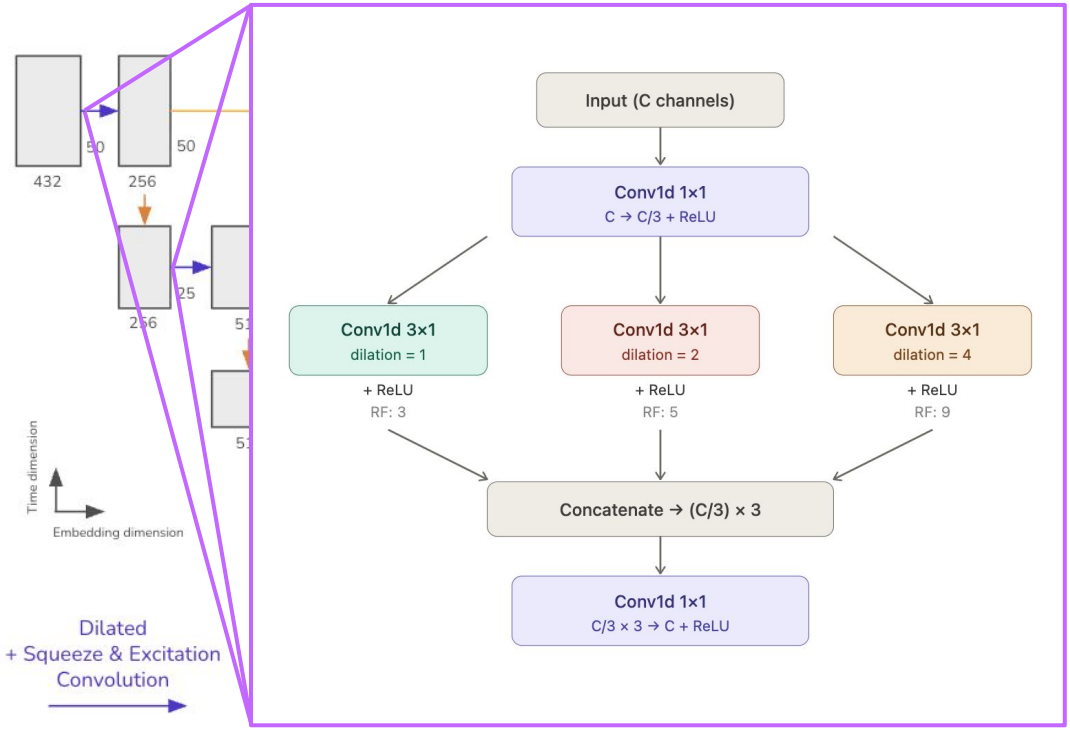


+0.43 (+1.3%)
in mAP10

Comparison with previous stage

- Very small impact on performance.
- Small differences in performance among the different kernels.

Dilated convolutions



2. F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," in Proc. Int. Conf. Learn. Represent. (ICLR), 2016.

+1.09 (+3%)
in mAP10

Comparison with previous stage

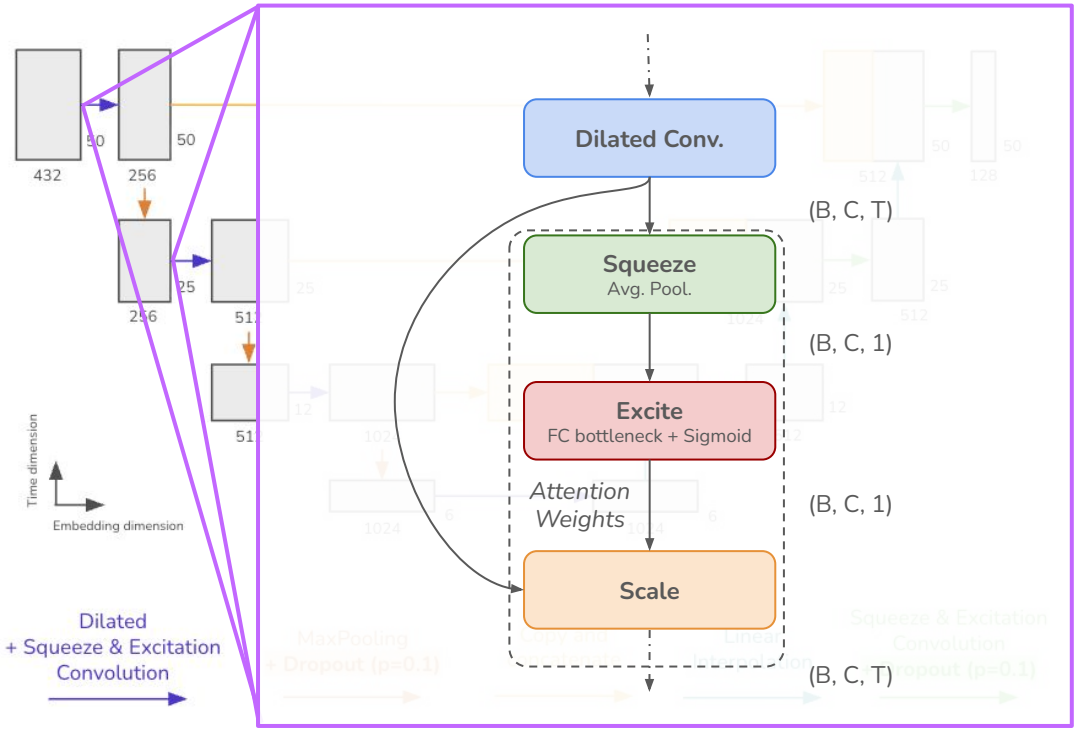
Problem:

Actions in videos happen at different time scales.
- Many stacked layers (depth)
- Larger kernels

Efficient solution:

Dilated convs. with 3 parallel branches in the encoder:
- RFs of 3, 5 and 9 with kernel size 3.
- Each branch specialized in a different temporal resolution.

Lightweight attention: Squeeze and Excitation blocks



3. Jie Hu, Li Shen, Gang Sun; Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2018, pp. 7132-7141

+0.63 (+2%)
in mAP10

Comparison with previous stage

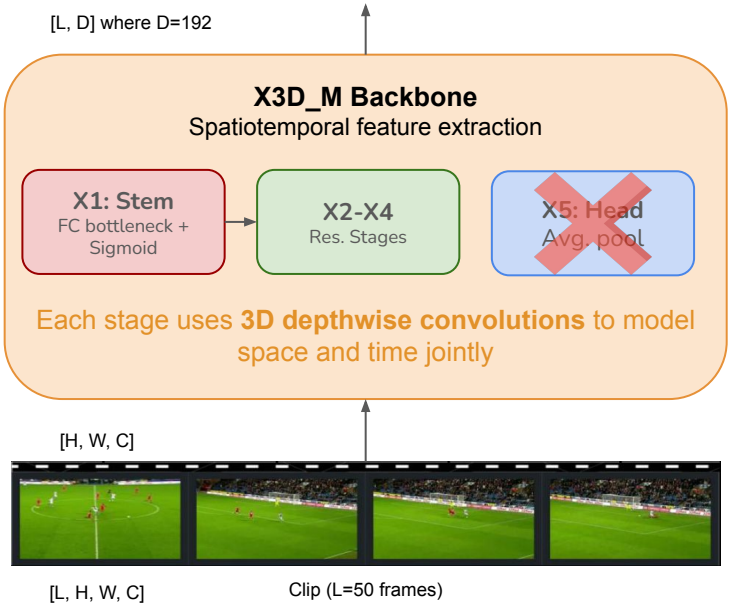
Problem:

After convolution, all channels are weighted equally — but not all features are relevant for every action

Solution:

Squeeze and Excitation blocks which learn channel importance and weight the channels.

Spatiotemporal backbone: X3D



+6.05 (+18%)
in mAP10

Comparison with previous stage

Problem:

2D backbone (RegNetY) processes each frame independently — temporal context is captured only in the U-Net head, not in the backbone itself.

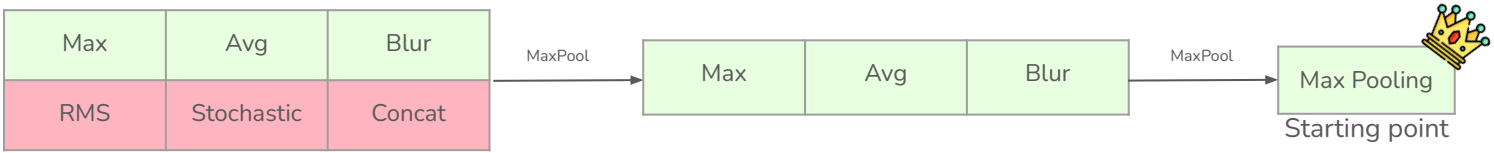
Solution:

X3D uses 3D convolutions jointly over space and time in the backbone itself, learning spatiotemporal features directly during extraction.

4. Feichtenhofer, Christoph. "X3D: Expanding Architectures for Efficient Video Understanding." CVPR, 2020.

Different Poolings

- **MaxPooling**: Take the maximum
- **AvgPooling**: Take the average
- **BlurPooling**[4]: MaxPool(s=1), then depthwise gaussian blur, finally dowsnample(s=2)
- **LPPooling (p=2)**: Square root of the mean
- **StochasticPooling**[5]: sample for each window proportional to the softmax of the activations; on inference uses a weighted avg
- **ConcatPooling**: MaxPool1d and AvgPool1d, doubles the number of channels

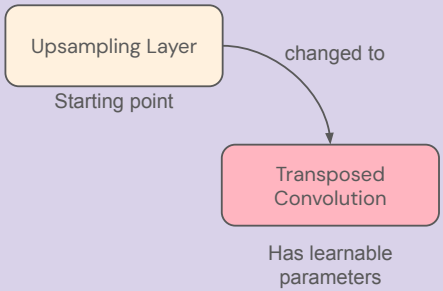


4. Zhang, Richard. *"Making Convolutional Networks Shift-Invariant Again."* Proceedings of the International Conference on Machine Learning (ICML), 2019.
5. Zeiler, Matthew D., and Rob Fergus. *"Stochastic Pooling for Regularization of Deep Convolutional Neural Networks."* arXiv:1301.3557, 2013.

3.2 Unsuccessful approaches ❌

Other approaches:

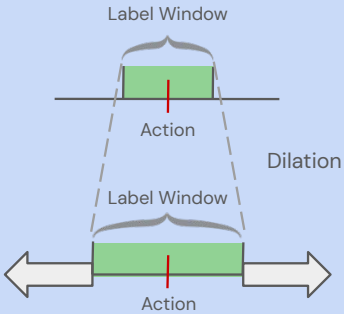
Transposed Convolution



mAP: **-1.3** ↓

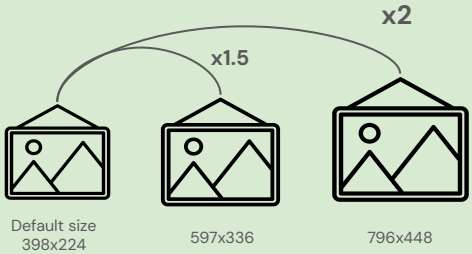
More Parameters !!

Label Dilation



mAP ↓

Increase Frame Size

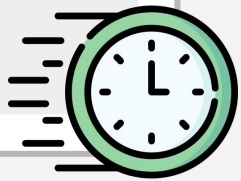


mAP: **-2.27** ↓ **-17.2** ↓↓

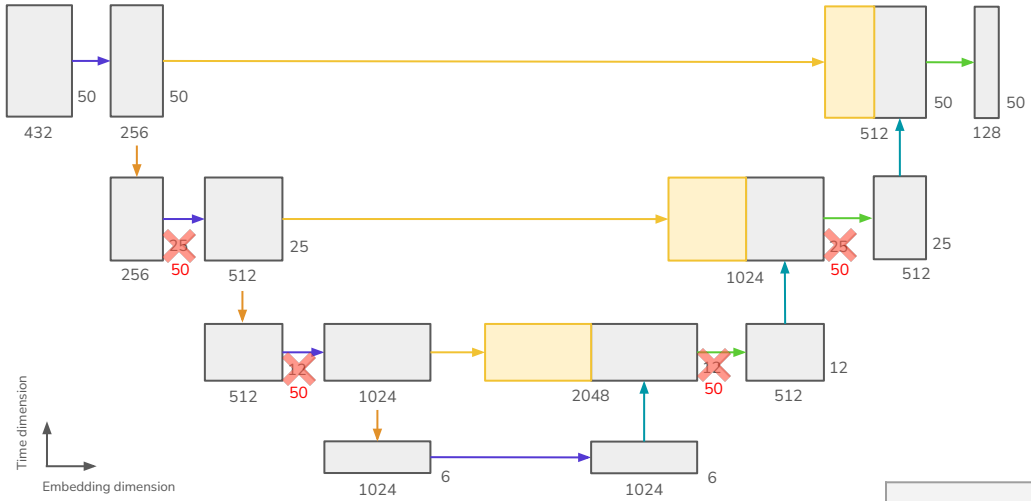
Time & Resources ↑↑

4

Impact of time aggregation



4 Impact of time aggregation



- Perform a fair comparison by **maintaining #params**
- **Removing** the temporal resolution **bottleneck**
  - Removing poolings and upscaling

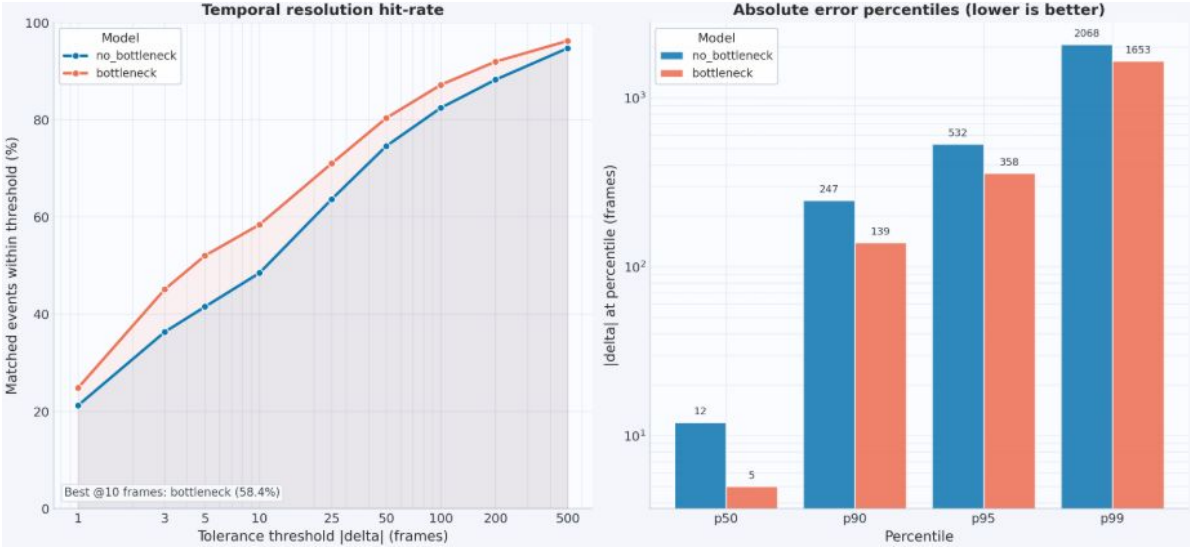
Comparison of the valuation at multiple tolerances **0.5s vs 1s**

- Reducing temporal resolution outperforms in all the mAP computations
- Counterintuitively, the bigger the tolerance, the smaller the gap

Target	Reducing temporal resolution	Maintaining temporal resoulution
mAP10 1s	40.14 (+1.66)	38.48
mAP10 0.5s	36.77 (+3.34)	33.36
mAP12 1s	35.73 (+2.38)	33.35
mAP12 0.5s	32.92 (+3.82)	29.10

4 Impact of time aggregation

Analysis by frame distance wrt GT



- Computed the % of matched events within a frame distance
- **Bottlenecking** the temporal resolution **outperformed** in all the tolerances ✅
- This behaviour might be explained due to X3D and the neck bottleneck **complementing** each other 🤔

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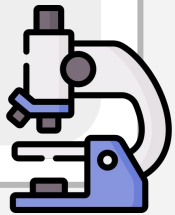
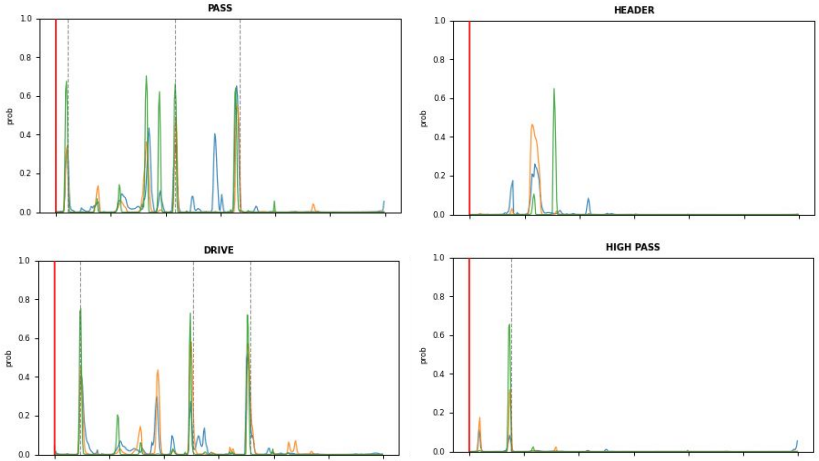
Further model analysis



5 Further model analysis

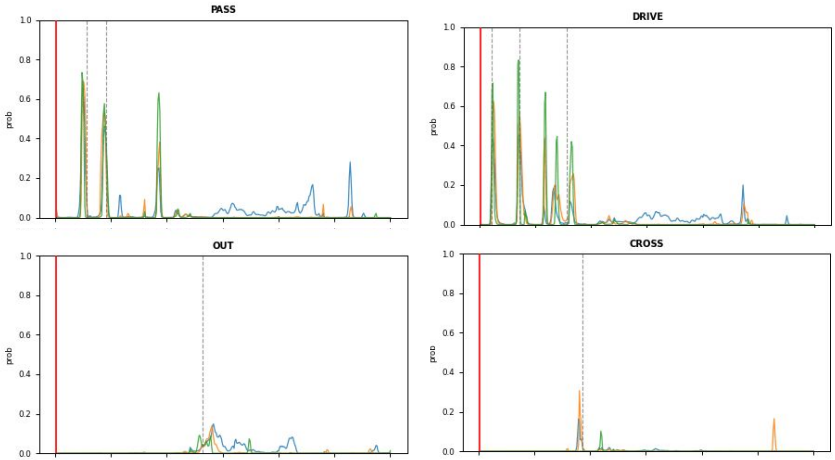


- **Ground truth gaps:** Some annotations have missing labels
- **Class generalization:** The model learned to reliably detect the header class even where GT labels are absent.

  Inter-class ambiguity: Visually similar actions cause the model to over-predict overlapping classes.
- **Prediction sharpness:** The baseline produces noisy, low-confidence outputs across time. Adding SE and the X3D backbone yields sharper, more peaked confidence curves.

Baseline Squeeze and Excite X3D

5 Further model analysis



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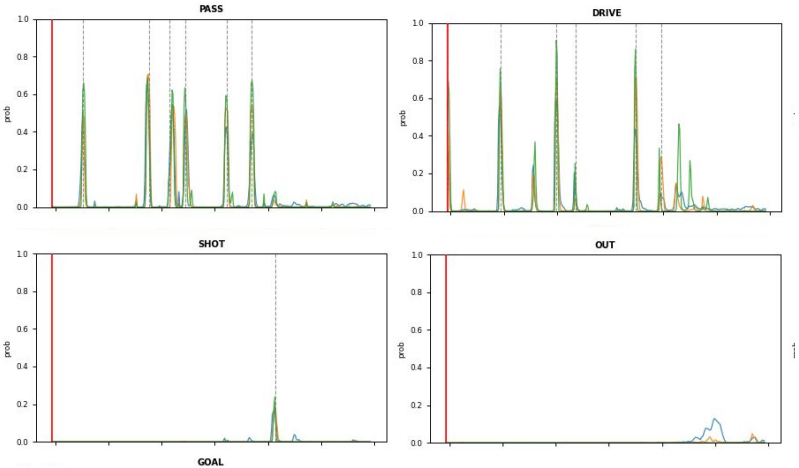

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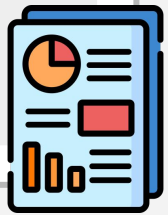
Baseline Squeeze and Excite X3D

6

Discussion & conclusions



6 Discussion & conclusions

KEY TAKEAWAYS

- A strong spatio-temporal backbone is the single most influential factor.
- Time aggregation was not a decisive factor. Maybe due to strong backbone.
- Hard to predict the influence of modifications.

CHALLENGES

Dataset:

- Severe class imbalance.
- Ground truth is not perfect.
- Visually similar classes.

Tasks:

- Both require large datasets and models → compute cost limits.
- Space of possible architectures is huge. Many different approaches. Hard to design efficient experiments.

FUTURE WORK

- Many hyperparameters have not been optimized. Dedicate more resources to this.
- Add attention gates on skip connections to suppress irrelevant temporal regions.

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